The Mean Value Theorems

Rolle, Lagrange and Cauchy

Engineering Mathematics–I (CSE-1001)
BTech 1st Year, IIIT Bhopal

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To the Student

This note opens **Module 2**, and it is the bridge between the single-variable calculus of Module 1 and the multivariable calculus that follows. You have already met one mean value theorem without being told its name: the **Extended Mean Value Theorem** used to prove Taylor's Theorem in the companion note *Approximation and Taylor's Theorem*. This note goes back to the beginning and builds the family properly — Rolle first, then Lagrange, then Cauchy — with a proof of each.

Everything here rests on a single picture: *somewhere between two points, the curve must do on the inside what the chord does on the outside.* If you hold that picture in mind, every theorem in this note is the same statement written with different amounts of generality.

How to use this note:

1. Attend the lecture for the topic, then read the matching section here, working each example yourself before reading its solution.
2. Attempt the matching questions in the companion question bank once a section is finished, and check the key only after a serious attempt.

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1 Average Rate Against Instantaneous Rate

Suppose you drive from Bhopal to Indore, a distance of 190 km, and the journey takes exactly two hours. Your **average** speed was

$$\frac{190 \text{ km}}{2 \text{ h}} = 95 \text{ km/h}.$$

Now here is the question that this entire module answers: *at some instant during the journey, was your speedometer reading exactly* 95?

It feels as though it must have been. You cannot average 95 without ever being at 95 — unless you somehow jumped from 80 to 110 without passing through the values in between. And a car cannot do that, because its speed varies continuously.

That intuition is correct, and the Mean Value Theorem is the precise statement of it.

The idea

The average rate of change of f over $[a, b]$ is the slope of the **chord** joining the endpoints:

$$\frac{f(b) - f(a)}{b - a}.$$

The instantaneous rate of change at a point c is the slope of the **tangent** there, $f'(c)$. The Mean Value Theorem says these two are equal *somewhere*: there is at least one interior point at which the tangent is parallel to the chord.

The speeding ticket you cannot argue with

Several highway authorities now issue tickets by **average speed enforcement**: cameras record the time you enter and leave a stretch of road, and your average speed over it is computed. If that average exceeds the limit, you are fined — and you cannot defend yourself by claiming you were never *actually* speeding at any instant. The Mean Value Theorem says you were. At some moment your speedometer read exactly the average, and if the average was above the limit, so was that reading.

2 Preliminaries: Intervals and Hypotheses

Every theorem in this note has hypotheses of the same shape:

*f is continuous on the **closed** interval $[a, b]$ and differentiable on the **open** interval (a, b).*

Students often ask why the two intervals are different. The answer is that each condition is asked for exactly where it is needed, and nowhere else.

- **Continuity is asked on** $[a, b]$ — including the endpoints — because the conclusion compares the values $f(a)$ and $f(b)$. Those two numbers must mean something, so f has to be well-behaved right up to the ends.
- **Differentiability is asked only on** (a, b) because the conclusion produces a point c *strictly inside*. We never need a derivative at a or at b, so we do not demand one.

This is not pedantry; it buys real generality. The function $f(x) = \sqrt{1 - x^2}$ on $[-1, 1]$ is continuous on the closed interval but has no derivative at either endpoint (the tangents there are vertical). Because the theorems only ask for differentiability inside, they still apply to it.

Notation for intervals

$[a, b]$ includes both endpoints; (a, b) excludes both; $[a, b)$ and $(a, b]$ include one each. An infinite end never takes a square bracket — we write $(-\infty, 5]$ and $(2, \infty)$, never $[-\infty, 5]$, because ∞ is not a number and cannot be "included".

3 Rolle's Theorem

Start with the special case in which the chord is horizontal.

Theorem 3.1 (Rolle's Theorem)**.** *Let $f : [a, b] \to \mathbb{R}$ satisfy*
(i) f is continuous on $[a, b]$;
(ii) f is differentiable on (a, b);
(iii) $f(a) = f(b)$.
Then there exists at least one $c \in (a, b)$ with

$$f'(c) = 0.$$

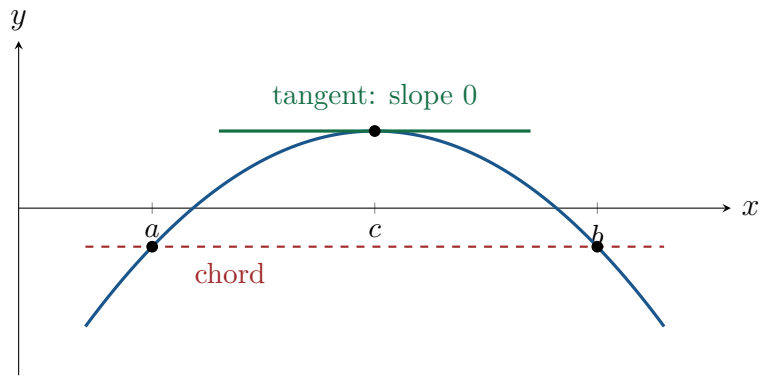


Figure 1: Rolle's Theorem. The chord is horizontal because $f(a) = f(b)$, and somewhere inside the tangent is horizontal too.

3.1 Proof

The proof is short, and it uses one fact from Module 1 which we take as given:

Extreme Value Theorem. A function continuous on a closed bounded interval $[a, b]$ attains both a maximum and a minimum value on $[a, b]$.

Proof of Rolle's Theorem. By the Extreme Value Theorem, f attains a maximum value M and a minimum value m on $[a, b]$. Two cases arise.

Case 1: $M = m$**.** Then f is constant on $[a, b]$, so $f'(x) = 0$ at *every* interior point, and any $c \in (a, b)$ will do.

Case 2: $M > m$**.** Since $f(a) = f(b)$, at least one of M and m differs from this common endpoint value, and is therefore attained at some interior point $c \in (a, b)$. Suppose it is the maximum (the minimum case is identical with inequalities reversed).

At this interior maximum, consider the two one-sided limits of the difference quotient. For small $h > 0$ we have $f(c + h) \leq f(c)$, so

$$\frac{f(c+h) - f(c)}{h} \leq 0 \qquad \Longrightarrow \qquad f'(c) = \lim_{h \to 0^+} \frac{f(c+h) - f(c)}{h} \leq 0.$$

For small $h < 0$ we still have $f(c + h) \leq f(c)$, but now h is negative, so the quotient is ≥ 0 and

$$f'(c) = \lim_{h \to 0^-} \frac{f(c+h) - f(c)}{h} \geq 0.$$

Both limits equal $f'(c)$, because f is differentiable at c. Hence $f'(c) \leq 0$ and $f'(c) \geq 0$ simultaneously, which forces

$$f'(c) = 0.$$

□

What the proof really says

At an interior maximum the function cannot be going up (there is no room above) and at an interior minimum the function cannot be going down (then a point just before would be higher). The only remaining possibility is that it is going nowhere: the derivative is zero. This is the whole content of the theorem, and it is why the point must be *interior* — at an endpoint the function is free to arrive still climbing.

3.2 A worked example

Example 3.2. Verify Rolle's Theorem for $f(x) = x^2 - 4x + 3$ on $[1, 3]$.

Solution. **Check the hypotheses.** f is a polynomial, hence continuous on $[1, 3]$ and differentiable on $(1, 3)$. Also

$$f(1) = 1 - 4 + 3 = 0, \qquad f(3) = 9 - 12 + 3 = 0,$$

so $f(1) = f(3)$. All three hypotheses hold.

Find c**.** $f'(x) = 2x - 4$, and $f'(c) = 0$ gives $c = 2$, which lies in $(1, 3)$. Rolle's Theorem is verified, and here the point c is unique. □

3.3 Every hypothesis is needed

A theorem's hypotheses are not decoration. Drop any one of the three and the conclusion fails.

Function	**Which hypothesis fails**	**Why no c exists**
$f(x) = \lvert x\rvert$ on $[-1, 1]$	(ii): no derivative at 0	$f'(x) = \pm 1$ everywhere else
$f(x) = x$ on $[0, 1]$	(iii): $f(0) \neq f(1)$	$f'(x) = 1$ for every x
$f(x) = 1 - (x-1)^{2/3}$ on $[0, 2]$	(ii): no derivative at 1	f' never vanishes on $(0, 2)$

In the first and third rows the function is perfectly continuous and the endpoint values do match — it is the single missing derivative that destroys the conclusion. The graph has a corner or a cusp exactly where the horizontal tangent ought to have been.

Try it yourself

Does Rolle's Theorem apply to $f(x) = \dfrac{\sin x}{e^x}$ on $[0, \pi]$? If so, find a value of c.

Answer: yes — $f(0) = f(\pi) = 0$ and f is differentiable everywhere. $f'(x) = e^{-x}(\cos x - \sin x) = 0$ gives $\tan c = 1$, so $c = \pi/4$.

4 Lagrange's Mean Value Theorem

Now tilt the picture. If the chord is not horizontal, the tangent should be parallel to it rather than flat.

Theorem 4.1 (Lagrange's Mean Value Theorem)**.** *Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). Then there exists at least one $c \in (a, b)$ with*

$$f'(c) \;=\; \frac{f(b) - f(a)}{b - a}, \qquad \textit{equivalently} \qquad f(b) - f(a) = f'(c)\,(b - a).$$

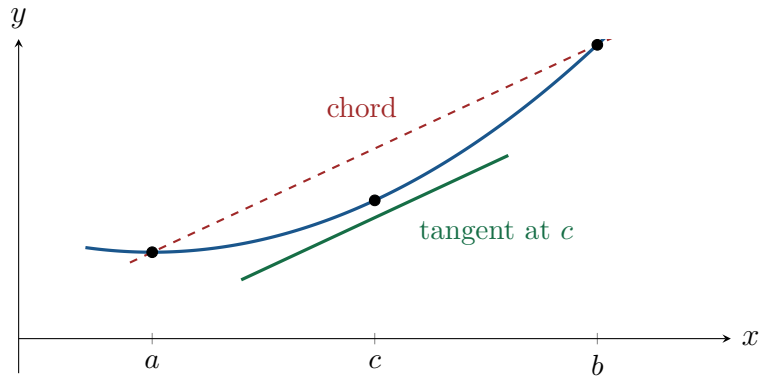


Figure 2: Lagrange's Mean Value Theorem. Somewhere inside, the tangent is parallel to the chord.

4.1 Proof: tilt the picture until Rolle applies

Rolle's Theorem handles a horizontal chord. Lagrange's does not look horizontal — so we subtract the chord off, which makes it so.

Proof. Let $L(x)$ be the straight line through the two endpoints,

$$L(x) = f(a) + \frac{f(b) - f(a)}{b - a}\,(x - a),$$

and define the **auxiliary function**

$$g(x) = f(x) - L(x).$$

Geometrically, $g(x)$ is the vertical gap between the curve and its chord.

g **satisfies the hypotheses of Rolle's Theorem.** It is continuous on $[a, b]$ and differentiable on (a, b), being a difference of two such functions. Moreover

$$g(a) = f(a) - L(a) = f(a) - f(a) = 0, \qquad g(b) = f(b) - L(b) = f(b) - f(b) = 0,$$

so $g(a) = g(b)$. (The chord meets the curve at both ends — that is what a chord is.)

Apply Rolle. There exists $c \in (a, b)$ with $g'(c) = 0$. But

$$g'(x) = f'(x) - \frac{f(b) - f(a)}{b - a},$$

since the slope of L is the constant $\frac{f(b)-f(a)}{b-a}$. Setting $g'(c) = 0$ gives

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$ □

Rolle is the special case

Put $f(a) = f(b)$ in Theorem 4.1 and the right-hand side becomes 0, returning Rolle's Theorem exactly. So the two theorems are not independent results: Lagrange's is the general statement, Rolle's is the tilted-flat case, and we proved the general one *from* the special one by removing the tilt.

Example 4.2. Verify Lagrange's Mean Value Theorem for $f(x) = \sin x$ on $[0, \pi]$.

Solution. f is continuous and differentiable everywhere. The chord slope is

$$\frac{f(\pi) - f(0)}{\pi - 0} = \frac{0 - 0}{\pi} = 0,$$

so we need $f'(c) = \cos c = 0$ with $c \in (0, \pi)$. This gives $c = \pi/2$. The theorem is verified.

(Notice that $f(0) = f(\pi)$ here, so this is really an instance of Rolle's Theorem in disguise.) □

4.2 Three consequences you will use constantly

Lagrange's theorem is the engine behind several facts you have been using since school without proof.

Corollary 4.3 (Zero derivative means constant)**.** *If $f'(x) = 0$ for every x in an interval I, then f is constant on I.*

Proof. Take any two points $x_1 < x_2$ in I. By Theorem 4.1 applied on $[x_1, x_2]$, there is c between them with

$$f(x_2) - f(x_1) = f'(c)(x_2 - x_1) = 0 \cdot (x_2 - x_1) = 0,$$

so $f(x_2) = f(x_1)$. Since the two points were arbitrary, f takes the same value everywhere on I. □

Corollary 4.4 (Sign of the derivative controls monotonicity)**.** *If $f'(x) > 0$ throughout an interval then f is strictly increasing there; if $f'(x) < 0$ throughout, f is strictly decreasing.*

Proof. For $x_1 < x_2$, $f(x_2) - f(x_1) = f'(c)(x_2 - x_1)$. The bracket is positive, so the sign of the left side is the sign of $f'(c)$. □

Corollary 4.5 (Bounding a function by its derivative)**.** *If $|f'(x)| \le M$ for all $x \in (a, b)$, then*

$$|f(x_2) - f(x_1)| \le M\,|x_2 - x_1| \qquad \text{for all } x_1, x_2 \in [a, b].$$

This last one is how inequalities get proved, and we use it in Section 6.

"There is a c" does not mean "I know which c"

The Mean Value Theorem is an **existence** theorem. It guarantees that some suitable c exists; it does not tell you where, and in general c cannot be found in closed form. In examples we can often solve for it because the functions are simple — but a proof that *needs* to know the value of c is almost always a wrong proof. What legitimate proofs use is only this: c lies between a and b, so anything true for all points of that interval is true at c.

Try it yourself

Use Corollary 4.5 to show that $|\sin x - \sin y| \le |x - y|$ for all real x and y.

Answer: apply the MVT to sin *on the interval between x and y:* $\sin x - \sin y = \cos(c)(x - y)$ *for some c. Since* $|\cos c| \le 1$*, taking absolute values gives the result.*

5 Cauchy's Mean Value Theorem

The last member of the family compares *two* functions on the same interval.

Theorem 5.1 (Cauchy's Mean Value Theorem)**.** *Let f and g be continuous on $[a, b]$ and differentiable on (a, b), with $g'(x) \neq 0$ for every $x \in (a, b)$. Then there exists $c \in (a, b)$ with*

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}.$$

Why $g(b) \neq g(a)$, so the left side makes sense

If $g(b)$ equalled $g(a)$, Rolle's Theorem applied to g would produce a point where $g' = 0$ — contradicting the hypothesis $g' \neq 0$ on (a, b). So the denominator on the left cannot vanish. This is a small point, but a careful proof must make it.

Proof. Define the auxiliary function

$$h(x) = \Big[f(b) - f(a)\Big]\, g(x) \;-\; \Big[g(b) - g(a)\Big]\, f(x).$$

It is continuous on $[a, b]$ and differentiable on (a, b). Evaluating at the endpoints:

$$h(a) = f(b)g(a) - f(a)g(a) - g(b)f(a) + g(a)f(a) = f(b)g(a) - g(b)f(a),$$

$$h(b) = f(b)g(b) - f(a)g(b) - g(b)f(b) + g(a)f(b) = g(a)f(b) - f(a)g(b).$$

These are equal, so $h(a) = h(b)$ and Rolle's Theorem gives a point $c \in (a, b)$ with $h'(c) = 0$, that is

$$\Big[f(b) - f(a)\Big]g'(c) = \Big[g(b) - g(a)\Big]f'(c).$$

Dividing by $g'(c)\Big[g(b) - g(a)\Big]$, both non-zero, gives the result. □

The most common wrong proof in this module

Many students "prove" Cauchy's theorem like this:

> *By the Mean Value Theorem, $f(b) - f(a) = f'(c)(b - a)$ and $g(b) - g(a) = g'(c)(b - a)$. Dividing one by the other gives the result.*

This is **wrong**. The Mean Value Theorem applied to f produces some point c_1, and applied to g produces some point c_2, and *there is no reason whatever for c_1 and c_2 to be the same point.* What the argument actually establishes is

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c_1)}{g'(c_2)},$$

which is a far weaker statement and useless for everything Cauchy's theorem is used for. The whole difficulty of the theorem — and the reason it needs its own auxiliary function — is forcing a *single* c to serve both functions at once.

Lagrange is the special case

Put $g(x) = x$ in Theorem 5.1. Then $g'(x) = 1 \neq 0$, $g(b) - g(a) = b - a$, and the conclusion becomes $\frac{f(b)-f(a)}{b-a} = f'(c)$ — Lagrange's theorem. The three results therefore nest:

$$\text{Rolle} \subset \text{Lagrange} \subset \text{Cauchy},$$

each proved from the one before it by a well-chosen auxiliary function.

Example 5.2. Verify Cauchy's Mean Value Theorem for $f(x) = x^2$ and $g(x) = x^3$ on $[1, 2]$.

Solution. Both are polynomials, so continuous on $[1,2]$ and differentiable on $(1,2)$; and $g'(x) = 3x^2 \neq 0$ on $(1,2)$. The left-hand side is

$$\frac{f(2)-f(1)}{g(2)-g(1)} = \frac{4-1}{8-1} = \frac{3}{7}.$$

The right-hand side is

$$\frac{f'(c)}{g'(c)} = \frac{2c}{3c^2} = \frac{2}{3c}.$$

Setting these equal, $\dfrac{2}{3c} = \dfrac{3}{7}$ gives $c = \dfrac{14}{9} \approx 1.556$, which lies in $(1,2)$. The theorem is verified. □

Try it yourself

Verify Cauchy's Mean Value Theorem for $f(x) = \sqrt{x}$ and $g(x) = \dfrac{1}{\sqrt{x}}$ on $[1,3]$.

Answer: the left side is $\frac{\sqrt{3}-1}{\frac{1}{\sqrt{3}}-1} = -\sqrt{3}$. *The right side is* $\frac{1/(2\sqrt{c})}{-1/(2c^{3/2})} = -c$. *Hence* $c = \sqrt{3} \approx 1.732$, *which lies in* $(1,3)$.

6 Applications

The mean value theorems are rarely the *answer* to a question. They are the tool that turns a statement about values of f into a statement about f', or the reverse. Three patterns cover most examination questions.

6.1 Pattern 1 — locating roots of the derivative

Proposition 6.1. *Between any two consecutive real roots of a polynomial P there lies at least one real root of P'.*

Proof. Let $r_1 < r_2$ be roots of P. Then P is continuous on $[r_1, r_2]$, differentiable on (r_1, r_2), and $P(r_1) = 0 = P(r_2)$. Rolle's Theorem gives a $c \in (r_1, r_2)$ with $P'(c) = 0$. □

Example 6.2. Illustrate Proposition 6.1 for $P(x) = (x-1)(x-3)(x+2)$.

Solution. The roots of P are -2, 1 and 3, so there are two consecutive gaps: $(-2,1)$ and $(1,3)$. Expanding, $P(x) = x^3 - 2x^2 - 5x + 6$, so

$$P'(x) = 3x^2 - 4x - 5, \qquad x = \frac{4 \pm \sqrt{16+60}}{6} = \frac{4 \pm \sqrt{76}}{6}.$$

Numerically the roots of P' are

$$x \approx -0.7863 \quad \text{and} \quad x \approx 2.1196,$$

and indeed $-0.7863 \in (-2,1)$ and $2.1196 \in (1,3)$ — exactly one in each gap, as the proposition promises. □

6.2 Pattern 2 — counting roots by monotonicity

Rolle's Theorem run backwards limits how *many* roots a function can have: if f' never vanishes on an interval, then f cannot have two roots there (two roots would force a vanishing derivative in between).

Example 6.3. Show that the equation

$$(x-1)^3 + (x-2)^3 + (x-4)^3 + (x-5)^3 = 0$$

has exactly one real root.

Solution. Call the left-hand side $f(x)$.

At least one root. Expanding, the leading term is $4x^3$, so f is a cubic polynomial; every cubic has at least one real root (it tends to $-\infty$ at one end and $+\infty$ at the other, and is continuous).

At most one root. Differentiating,

$$f'(x) = 3\Big[(x-1)^2 + (x-2)^2 + (x-4)^2 + (x-5)^2\Big].$$

This is a sum of squares, and the four squares cannot vanish simultaneously (they would need x to equal 1, 2, 4 and 5 at once). Hence $f'(x) > 0$ for every real x, so f is strictly increasing and cannot take the value 0 twice.

Therefore the equation has **exactly one** real root. (In fact the four numbers $1, 2, 4, 5$ are symmetric about 3, and substituting $x = 3$ gives $8 + 1 - 1 - 8 = 0$, so the root is exactly $x = 3$.) □

Check your derivative against the equation you were given

A frequent slip in this type of question is to differentiate a *different* expression from the one in the problem — for example writing $3[(x-1)^2+(x-2)^2+(x-3)^2+(x-4)^2]$ above, with 3 and 4 in place of 4 and 5. The argument survives, because any sum of squares is positive, but the derivative does not belong to the stated equation and the work is wrong. Always read your own f' back against the original f.

6.3 Pattern 3 — proving inequalities

Corollary 4.5 converts a bound on f' into a bound on the change in f. Most inequality questions in this module are that corollary in disguise.

Example 6.4. Show that for all $x, y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$,

$$|\tan x - \tan y| \;\geq\; |x - y|\,.$$

Solution. Assume $x \neq y$ (otherwise both sides are 0), and apply Theorem 4.1 to $f(t) = \tan t$ on the interval between y and x. The function is continuous and differentiable throughout $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, so there is a c strictly between them with

$$\tan x - \tan y = \sec^2(c)\,(x - y).$$

Now $\sec^2 c = 1 + \tan^2 c \geq 1$ for every real c. Taking absolute values,

$$|\tan x - \tan y| = \sec^2(c)\,|x - y| \geq |x - y|\,.$$ □

Compare the two inequalities

Section 4 gave $|\sin x - \sin y| \le |x - y|$ and this section gives $|\tan x - \tan y| \ge |x - y|$ — opposite directions, identical method. The direction is decided entirely by whether the derivative is bounded *above* by 1 ($|\cos c| \le 1$) or *below* by 1 ($\sec^2 c \ge 1$). Identify the derivative, bound it, and the inequality writes itself.

Example 6.5. Show that $e^x \ge 1 + x$ for every real x.

Solution. Fix $x \neq 0$ and apply the Mean Value Theorem to $f(t) = e^t$ on the interval between 0 and x: there is a c strictly between them with

$$e^x - e^0 = e^c(x - 0), \qquad \text{that is} \qquad e^x = 1 + x\,e^c.$$

If $x > 0$ then $c > 0$, so $e^c > 1$ and $xe^c > x$. If $x < 0$ then $c < 0$, so $e^c < 1$, and multiplying by the *negative* number x reverses the inequality to give $xe^c > x$ again. Either way $e^x = 1 + xe^c \ge 1 + x$, with equality only at $x = 0$. □

Try it yourself

Use the Mean Value Theorem to show that for $0 < a < b$,

$$\frac{b-a}{b} < \ln\frac{b}{a} < \frac{b-a}{a}.$$

Hint: apply the theorem to $f(t) = \ln t$ on $[a, b]$ and bound $f'(c) = 1/c$ using $a < c < b$.

7 Summary

Theorem	Extra hypothesis	Conclusion
Rolle	$f(a) = f(b)$	$f'(c) = 0$
Lagrange	—	$f'(c) = \dfrac{f(b) - f(a)}{b - a}$
Cauchy	second function g, $g' \neq 0$	$\dfrac{f'(c)}{g'(c)} = \dfrac{f(b) - f(a)}{g(b) - g(a)}$

All three share the hypotheses *continuous on* $[a, b]$, *differentiable on* (a, b), all three conclude with *there exists* $c \in (a, b)$, and each is proved from the one above it by subtracting a cleverly chosen auxiliary function so that Rolle's Theorem applies.

The one sentence to remember

Somewhere strictly inside the interval, the curve does on the inside what the chord does on the outside.

Looking ahead. In the companion note *Approximation and Taylor's Theorem* you met the *Extended* Mean Value Theorem, which carries this idea to higher derivatives and yields Taylor's Theorem with the Lagrange remainder. In the next note of Module 2 we

leave one variable behind and ask what "derivative" should even mean for a function of two variables — and the first surprise is that the obvious answer is not good enough.

Note: After completing these notes, proceed to the ***companion Question Bank*** *for self-assessment.*

Best wishes.
